MODULE 08 · DOMAIN 2 - DATA PREPARATION

Chunking: size, overlap and arithmetic

Two limits every chunk must respect, the trade-offs of size and overlap, and the sums the exam asks you to do.

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Chunking is slicing a loaf. Too thick and nobody can take a bite. Too thin and every slice falls apart.

Three reasons documents are split

- **The embedding model has a window.** Text beyond it is ignored.
- **The LLM has a context budget.** Several chunks, the prompt and the answer must all fit.
- **One vector should mean one thing.** A chunk covering five topics matches none of them well.
- Each chunk becomes **one row** in a table, and one vector in the index.

> ON THE EXAM
>
> A whole document squeezed into one vector is a blurred average. Retrieval needs sharp, focused vectors.

The hosted embedding models' windows

MODEL	WINDOW	DIMENSIONS	NOTE
databricks-bge-large-en	512 tokens	1,024	Short window: keep chunks small
databricks-gte-large-en	8,192 tokens	1,024	Long window, not normalised

ON THE EXAM

Databricks recommends chunks well under GTE's 8,192 limit, so a wider variety of passages fits into the LLM's prompt.

Every chunk must pass two checks

01

Check 1

Chunk size is within the embedding window.

02

Check 2

k chunks, plus prompt, plus answer, fit the LLM window.

03

Then

Choose size for meaning, inside both limits.

ON THE EXAM

The embedding window caps one chunk. The LLM window caps all the chunks you retrieve, times k.

The context budget

ITEM	TOKENS
LLM context window	16,384
System prompt and instructions	minus 600
User question	minus 100
Room reserved for the answer	minus 1,000
Left for retrieved chunks	14,684
So at 2,000-token chunks, k can be at most	7
At 1,000-token chunks, k can be at most	14

ON THE EXAM

Halving the chunk size doubles how many distinct passages the model can see. That is why smaller chunks are usually recommended.

Small chunks or large chunks?

SMALLER CHUNKS

Precise, but thin

- Sharp vectors, precise matches
- More passages fit the prompt
- Risk: meaning split across chunks
- More rows in the index

LARGER CHUNKS

Contextual, but blurred

- More surrounding context kept
- Fewer rows in the index
- Risk: vectors average many topics
- Fewer fit the prompt, higher cost

THE TIEBREAKER

Answers cut off mid-thought suggest chunks too small. Irrelevant material in every chunk suggests chunks too large.

How size and overlap change the record count

A 10,000-TOKEN DOCUMENT	STRIDE	CHUNKS
500 tokens, 100 overlap	400	25
1,000 tokens, 200 overlap	800	13
1,000 tokens, no overlap	1,000	10
2,000 tokens, 100 overlap	1,900	6

ON THE EXAM

Stride is chunk size minus overlap. Chunks are roughly document length divided by stride. Bigger chunks and less overlap mean fewer records.

More overlap or less?

MORE OVERLAP

Safer boundaries

- A sentence cut at an edge appears whole in the next chunk
- Better recall of ideas near edges
- More records and duplicate text
- Higher storage and embedding cost

LESS OVERLAP

Leaner index

- Fewer records
- Less duplicated text in results
- Risk: ideas split at boundaries
- Lower cost to embed and store

THE TIEBREAKER

Overlap is insurance against bad cuts. If you split on natural boundaries, you need less of it.

Ways to decide where to cut

STRATEGY	CUTS AT	BEST FOR
Fixed size	Every N tokens, with overlap	Uniform text with no structure
Recursive	Paragraphs, then sentences, then words	General prose. A sensible default
Structure-aware	Headings, sections, clauses, HTML tags	Manuals, policies, contracts, web pages
One record per chunk	Each FAQ pair, product or ticket	Collections of small, separate items
Advanced	Meaning, hierarchy, summaries	Module 09

ON THE EXAM

The more structure a document has, the more the cuts should follow it. Fixed size ignores structure entirely.

Make every chunk stand on its own

- **Prefix** each chunk with its document title and section heading.
- Keep a **table whole**, with its header row, in one chunk.
- Keep a **question with its answer**, and a clause with its conditions.
- Store source, section and page as **metadata** columns, for citations and filters.
- A chunk that says **it** without saying what it is will never be retrieved for the right question.

> ON THE EXAM
>
> A good chunk reads sensibly on its own. If a person could not understand it out of context, neither can the retriever.

How Harbourline chunks its policy wordings

01
Split
One chunk per clause, following section headings.

02
Prefix
Policy name, section and clause number.

03
Guard
Long clauses split recursively, with small overlap.

04
Tag
Product, year and page as metadata.

05
Check
Every chunk under the embedding window.

ON THE EXAM

Clause boundaries do the work that overlap would otherwise do. Only the long clauses need the insurance.

A rag application retrieves the top 10 chunks, each about 2,000 tokens, and sends them to an LLM with a 16,384-token context window.

Many requests now fail with an error saying the maximum context length has been exceeded.

Which change will fix the error?

A Use smaller chunks, retrieve fewer of them, or both

B Raise the temperature of the LLM

C Increase the overlap between chunks

D Switch to an embedding model with a longer window

◆ PAUSE AND DECIDE

Which change will fix the error?

A Smaller chunks, fewer chunks

Ten times 2,000 is 20,000 before the prompt. Shrink either factor.

B Higher temperature

Changes randomness. Does not change the token count.

C More overlap

Makes chunks larger and more repetitive. Worse.

D Longer embedding window

Wrong limit. The error is the LLM's window, not the embedding model's.

THE SIGNAL

Maximum context length exceeded is an LLM limit. It is fixed by k times chunk size, never by the embedding model.

> A team's AI Search index has grown to 40 million chunk records. Cost and indexing time are now a concern.
>
> The team must roughly halve the number of records without removing any source content.

Which change achieves this?

A Reduce the chunk size

B Increase the overlap between chunks

C Add more metadata columns to each record

D Increase the chunk size and reduce the overlap

◆ PAUSE AND DECIDE

Which change achieves this?

A. Smaller chunks

A shorter stride means more records, not fewer.

B. More overlap

Also shortens the stride. More records.

C. More metadata

Wider rows. The row count is unchanged.

D. Bigger chunks, less overlap

Both lengthen the stride, so fewer chunks per document.

THE SIGNAL

Records are roughly length divided by stride. Stride is size minus overlap. Grow the size, shrink the overlap.

> An engineer chunks technical manuals into sections of about 1,500 tokens and embeds them with databricks-bge-large-en.
>
> Questions about content near the start of each section retrieve well. Questions about content later in a section almost never do.

What is the most likely cause?

A BGE embeddings are normalised, which weakens later content

B BGE's 512-token window means the rest of each chunk is never embedded

C The overlap between chunks is too small

D The LLM's context window is too small for 1,500-token chunks

◆ PAUSE AND DECIDE

What is the most likely cause?

A Normalisation

Normalising changes vector length, not which text is read.

B 512-token window

Text past token 512 never reaches the vector. It cannot match.

C Overlap too small

Overlap affects edges between chunks, not the second half of each.

D LLM window

Retrieval fails before the LLM is involved.

THE SIGNAL

Start retrieves well, later content never does. A position-dependent failure points to the embedding window.

> A support team indexes an FAQ document containing 400 short question-and-answer pairs, using fixed 1,000-token chunks.
>
> Answers often combine the answer to one question with details from an unrelated one.

What should the engineer change?

A Increase the chunk size to 2,000 tokens for more context

B Increase the overlap between chunks to 300 tokens each

C Chunk by structure: one question-and-answer pair per chunk

D Switch to a larger embedding model with more dimensions

◆ PAUSE AND DECIDE

What should the engineer change?

A 2,000-token chunks

Packs even more unrelated pairs together.

B More overlap

Adds fragments of neighbouring pairs. More mixing.

C One pair per chunk

Follows the document's own structure. One idea per vector.

D Larger embedding model

A sharper model still embeds a mixed chunk.

THE SIGNAL

Four hundred short, separate items cut by a fixed size. When the document has natural units, chunk on them.

What to carry forward

- Two limits: **one chunk** within the embedding window, **k chunks** plus prompt and answer within the LLM window.
- BGE large: 512 tokens. GTE large: 8,192. Both 1,024 dimensions.
- Records roughly equal length divided by stride. Fewer records: **bigger chunks, less overlap**.
- Follow the document's structure: clauses, sections, one FAQ pair per chunk.
- Prefix chunks with title and heading. Keep tables whole.

> ON THE EXAM
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> Next: module 09 covers advanced chunking and re-ranking, for when simple splitting is not enough.